WORKSHEET 7.7

Exponential Growth and Decay

Unit 7: Differential Equations

Key ideas

- If $\dfrac{dy}{dt} = ky$, then $y = y_0 e^{kt}$, where y_0 is the value at $t = 0$.
- $k > 0$ is growth; $k < 0$ is decay.
- **Doubling time** $= \dfrac{\ln 2}{k}$; **half-life** $= \dfrac{\ln 2}{|k|}$. Neither depends on the starting amount.
- If the rate is proportional to the *difference* from a fixed level L, that difference decays exponentially: $y - L = (y_0 - L)e^{kt}$.

Worked examples

Example 1 · Growth

Solve $\dfrac{dy}{dt} = 0.08y$ with $y(0) = 250$, and find $y(10)$.

Solution. $y = 250e^{0.08t}$.

$y(10) = 250e^{0.8} \approx 556.4$**.**

Example 2 · Half-life

A substance has a half-life of 8 days. How much of a 100 mg sample remains after 20 days?

Solution. $100\left(\dfrac{1}{2}\right)^{20/8} = 100 \cdot 2^{-2.5}$.

About 17.7 mg.

Example 3 · Doubling time

A quantity grows according to $\dfrac{dy}{dt} = 0.035y$. Find its doubling time.

Solution. $\dfrac{\ln 2}{0.035}$.

About 19.8 time units.

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Practice

A calculator may be used for decimal answers on this page.

CORE

1. Solve $\frac{dy}{dt} = 2y$ with $y(0) = 5$.

2. Solve $\frac{dy}{dt} = -0.3y$ with $y(0) = 40$.

3. Find the doubling time for $\frac{dy}{dt} = 0.07y$.

4. Find the half-life for $\frac{dy}{dt} = -0.02y$.

EXAM LEVEL

5. A population grows exponentially from 3000 to 4500 in 5 years. Find k, and the population after 12 years.

6. A radioactive isotope has a half-life of 5730 years. What fraction of a sample remains after 2000 years?

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7. A quantity satisfies $\dfrac{dy}{dt} = ky$ with $y(1) = 4$ and $y(3) = 36$. Find $y(0)$.

8. A hot plate at 85 °C cools in a 25 °C room according to $\dfrac{dT}{dt} = k(T - 25)$. After 10 minutes it is at 55 °C. Find its temperature after 20 minutes.

Question 9 · Free-response practice

An investment of 2000 dollars grows so that its value A satisfies $\dfrac{dA}{dt} = 0.05A$, where t is in years.

(a) Find $A(t)$.

(b) Find the value of the investment after 10 years.

(c) Find the time at which the value reaches 5000 dollars, and the rate at which it is growing at that time.

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Worked solutions

1. $y = 5e^{2t}$.
2. $y = 40e^{-0.3t}$.
3. $\dfrac{\ln 2}{0.07} \approx 9.902$.
4. $\dfrac{\ln 2}{0.02} \approx 34.657$.
5. $e^{5k} = 1.5$, so $k = \dfrac{\ln 1.5}{5} \approx 0.0811$. $P(12) = 3000(1.5)^{12/5} \approx 7939$.
6. $2^{-2000/5730} \approx 0.785$.
7. $e^{2k} = 9$, so $e^k = 3$ and $y(0) = \dfrac{4}{3}$.
8. The difference $T - 25$ falls from 60 to 30 in 10 minutes: it halves. After 20 minutes it is 15, so $T = 40$ °C.
9. (a) $A(t) = 2000e^{0.05t}$.
   (b) $2000e^{0.5} \approx 3297.44$ dollars.
   (c) $e^{0.05t} = 2.5$, so $t = 20 \ln 2.5 \approx 18.326$ years. The rate is $0.05(5000) = 250$ dollars per year.

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